



DEPARTMENT OF COMPUTER SCIENCE
CYBER SECURITY

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DIGITAL FORENSICS LAB

VOLATILITY TOOL

Install Volatility 3

```
Session Actions Edit View Help
(kali@kali)-[~]
$ git clone https://github.com/volatilityfoundation/volatility3.git
Cloning into 'volatility3' ...
remote: Enumerating objects: 50518, done.
remote: Counting objects: 100% (815/815), done.
remote: Compressing objects: 100% (294/294), done.
remote: Total 50518 (delta 687), reused 522 (delta 521), pack-reused 49703 (from 2)
Receiving objects: 100% (50518/50518), 10.21 MiB | 79.00 KiB/s, done.
Resolving deltas: 100% (39174/39174), done.
(kali@kali)-[~]
$
```

```
(root@kali)-[/home/kali]
$ git clone https://github.com/volatilityfoundation/volatility3.git
fatal: destination path 'volatility3' already exists and is not an empty directory.
(root@kali)-[/home/kali]
$ cd volatility3
(root@kali)-[/home/kali/volatility3]
$ ls
API_CHANGES.md  CODING_STYLE.md  doc  MANIFEST.in  README.md  venv  vol.py  volshell.spec
CITATION.cff    development  LICENSE.txt  pyproject.toml  test  volatility3  volshell.py  vol.spec
```

- **Find Your Memory File**

Identifies the OS version, architecture, and kernel details of the memory dump. Always run this first to confirm the image is valid and readable.

```
(root@kali)-[/home/kali/volatility3]
$ find /home/kali -name "Investigation*" 2>/dev/null
(root@kali)-[/home/kali/volatility3]
$
```

Move into folder

After navigating into the nested folder on Desktop, ls -la confirmed both memory files are present. Investigation-1.vmem and Investigation-2.raw are each 536MB in size.

```
(root@kali)-[/home/kali/Desktop/Practical Investigation Memory Files_1615939758250 (1)]
$ cd "Practical Investigation Memory Files_1615939758250 (1)"
(root@kali)-[/home/kali/Desktop/Practical Investigation Memory Files_1615939758250 (1)/Practical Investigation Memory Files_1615939758250 (1)]
$ ls -la
total 1048592
drwxrwxr-x 2 kali kali 4096 May 23 12:01 .
drwxrwxr-x 3 kali kali 4096 May 23 12:01 ..
-rwxr-xr-x 1 kali kali 536870912 May 17 00:51 Investigation-1.vmem
-rwxr-xr-x 1 kali kali 536870912 May 17 00:51 Investigation-2.raw
```

- **Windows Info**

Gets basic information about the memory image including OS version, architecture and kernel details.

```
(root@kali) ~ - /home/kali/Desktop/Practical Investigation Memory Files_1615939758250 (1)/Practical Investigation Memory Files_1615939758250 (1)
python3 /home/kali/volatility3/vol.py -f Investigation-2.raw windows.info
Volatility 3 Framework 2.28.1
Progress: 100.00 PDB scanning finished
Variable Value
Kernel Base 0x804d7000
DTB 0x39000
Symbols file: ///home/kali/volatility3/volatility3/symbols/windows/ntoskrnl.pdb/423320282DB842E7BA2B0BFC86A84D75-2.json.xz
Is64Bit False
IsPAE False
layer_name 0 WindowsIntel
memory_layer 1 FileLayer
KdDebuggerDataBlock 0x8054cf60
NTBuildLab 2600.xpsp_sp3_qfe.130704-0421
CSDVersion 3
KdVersionBlock 0x8054cf38
Major/Minor 15.2600
MachineType 332
KeNumberProcessors 1
SystemTime 2017-05-12 21:26:32+00:00
NtSystemRoot C:\WINDOWS
NtProductType NtProductWinNt
NtMajorVersion 5
NtMinorVersion 1
PE MajorOperatingSystemVersion 5
PE MinorOperatingSystemVersion 1
PE Machine 332
PE TimeDateStamp Thu Jul 4 02:58:58 2013
```

• Process List

Lists all active running processes that were visible in Windows process list at time of capture.

```
(root@kali) ~ - /home/kali/Desktop/Practical Investigation Memory Files_1615939758250 (1)/Practical Investigation Memory Files_1615939758250 (1)
python3 /home/kali/volatility3/vol.py -f Investigation-2.raw windows.pslist
Volatility 3 Framework 2.28.1
Progress: 100.00 PDB scanning finished
PID PPID ImageFileName Offset(V) Threads Handles SessionId Wow64 CreateTime ExitTime File output
4 0 System 0x823c8830 51 244 N/A False N/A N/A Disabled
348 4 smss.exe 0x82169020 3 19 N/A False 2017-05-12 21:21:55.000000 UTC N/A Disabled
596 348 csrss.exe 0x82161da0 12 352 0 False 2017-05-12 21:22:00.000000 UTC N/A Disabled
620 348 winlogon.exe 0x8216e020 23 536 0 False 2017-05-12 21:22:01.000000 UTC N/A Disabled
664 620 services.exe 0x821937f0 15 265 0 False 2017-05-12 21:22:01.000000 UTC N/A Disabled
676 620 lsass.exe 0x82191658 23 353 0 False 2017-05-12 21:22:01.000000 UTC N/A Disabled
836 664 svchost.exe 0x8221a2c0 19 211 0 False 2017-05-12 21:22:02.000000 UTC N/A Disabled
904 664 svchost.exe 0x821b5230 9 227 0 False 2017-05-12 21:22:03.000000 UTC N/A Disabled
1024 664 svchost.exe 0x821af7e8 79 1366 0 False 2017-05-12 21:22:03.000000 UTC N/A Disabled
1084 664 svchost.exe 0x8203b7a8 6 72 0 False 2017-05-12 21:22:06.000000 UTC N/A Disabled
1152 664 svchost.exe 0x821bec78 10 173 0 False 2017-05-12 21:22:06.000000 UTC N/A Disabled
1484 664 spoolsv.exe 0x821e2da0 14 124 0 False 2017-05-12 21:22:09.000000 UTC N/A Disabled
1636 1608 explorer.exe 0x821d0da0 11 331 0 False 2017-05-12 21:22:10.000000 UTC N/A Disabled
1940 1636 tasksche.exe 0x82218da0 7 51 0 False 2017-05-12 21:22:14.000000 UTC N/A Disabled
1956 1636 ctfmon.exe 0x82231da0 1 86 0 False 2017-05-12 21:22:14.000000 UTC N/A Disabled
260 664 svchost.exe 0x81fb95d8 5 105 0 False 2017-05-12 21:22:18.000000 UTC N/A Disabled
740 1940 @WanaDecryptor@ 0x81fde308 2 70 0 False 2017-05-12 21:22:22.000000 UTC N/A Disabled
1768 1024 wuauclt.exe 0x81f747c0 7 132 0 False 2017-05-12 21:22:52.000000 UTC N/A Disabled
544 664 alg.exe 0x82010020 6 101 0 False 2017-05-12 21:22:55.000000 UTC N/A Disabled
1168 1024 wscntfy.exe 0x81fea8a0 1 37 0 False 2017-05-12 21:22:56.000000 UTC N/A Disabled
```

Process list revealed all running processes including the highly suspicious @WanaDecryptor@ (PID 740) clear sign of **WannaCry ransomware** infection.

```
740 1940 @WanaDecryptor@ 0x81fde308 2 70 0 False 2017-05-12 21:22:22.000000 UTC N/A Disabled
1768 1024 wuauclt.exe 0x81f747c0 7 132 0 False 2017-05-12 21:22:52.000000 UTC N/A Disabled
544 664 alg.exe 0x82010020 6 101 0 False 2017-05-12 21:22:55.000000 UTC N/A Disabled
```

• Process Scan

Scans raw memory for hidden or terminated processes that malware tries to conceal from pslist. The pslist shows @WanaDecryptor@ (PID 740), wuaucit.exe (PID 1768), and alg.exe (PID 544) as the last processes. This confirms WannaCry's decryptor GUI was actively running.

```
(root@kali) ~/home/kali/Desktop/Practical Investigation Memory Files_1615939758250 (1)/Practical Investigation Memory Files_1615939758250 (1)
python3 /home/kali/volatility3/vol.py -f Investigation-2.raw windows.psscan
```

Volatility 3 Framework 2.28.1
Progress: 100.00 PDB scanning finished

PID	PPID	ImageFileName	Offset(V)	Threads	Handles	SessionId	Wow64	CreateTime	ExitTime	File output
860	1940	taskdl.exe	0x1f4daf0	0	-	0	False	2017-05-12 21:26:23.000000 UTC	2017-05-12 21:26:23.000000 UTC	Disabled
536	1940	taskse.exe	0x1f53d18	0	-	0	False	2017-05-12 21:26:22.000000 UTC	2017-05-12 21:26:23.000000 UTC	Disabled
424	1940	@WanaDecryptor@	0x1f69b50	0	-	0	False	2017-05-12 21:25:52.000000 UTC	2017-05-12 21:25:53.000000 UTC	Disabled
1768	1024	wuaucit.exe	0x1f747c0	7	132	0	False	2017-05-12 21:22:52.000000 UTC	N/A	Disabled
576	1940	@WanaDecryptor@	0x1f8ba58	0	-	0	False	2017-05-12 21:26:22.000000 UTC	2017-05-12 21:26:23.000000 UTC	Disabled
260	664	svchost.exe	0x1fb95d8	5	105	0	False	2017-05-12 21:22:18.000000 UTC	N/A	Disabled
740	1940	@WanaDecryptor@	0x1fde308	2	70	0	False	2017-05-12 21:22:22.000000 UTC	N/A	Disabled
1168	1024	wscntfy.exe	0x1fea8a0	1	37	0	False	2017-05-12 21:22:56.000000 UTC	N/A	Disabled
544	664	alg.exe	0x2010020	6	101	0	False	2017-05-12 21:22:55.000000 UTC	N/A	Disabled
1084	664	svchost.exe	0x203b7a8	6	72	0	False	2017-05-12 21:22:03.000000 UTC	N/A	Disabled
596	348	csrss.exe	0x2161da0	12	352	0	False	2017-05-12 21:22:00.000000 UTC	N/A	Disabled
348	4	smss.exe	0x2169020	3	19	N/A	False	2017-05-12 21:21:55.000000 UTC	N/A	Disabled
620	348	winlogon.exe	0x216e020	23	536	0	False	2017-05-12 21:22:01.000000 UTC	N/A	Disabled
676	620	lsass.exe	0x2191658	23	353	0	False	2017-05-12 21:22:01.000000 UTC	N/A	Disabled
664	620	services.exe	0x21937f0	15	265	0	False	2017-05-12 21:22:01.000000 UTC	N/A	Disabled
1024	664	svchost.exe	0x21af7e8	79	1366	0	False	2017-05-12 21:22:03.000000 UTC	N/A	Disabled
904	664	svchost.exe	0x21b5230	9	227	0	False	2017-05-12 21:22:03.000000 UTC	N/A	Disabled
1152	664	svchost.exe	0x21bea78	10	173	0	False	2017-05-12 21:22:06.000000 UTC	N/A	Disabled
1636	1608	explorer.exe	0x21d9da0	11	331	0	False	2017-05-12 21:22:10.000000 UTC	N/A	Disabled
1484	664	spoolsv.exe	0x21e2da0	14	124	0	False	2017-05-12 21:22:09.000000 UTC	N/A	Disabled
1940	1636	tasksche.exe	0x2218da0	7	51	0	False	2017-05-12 21:22:14.000000 UTC	N/A	Disabled
836	664	svchost.exe	0x221a2c0	19	211	0	False	2017-05-12 21:22:02.000000 UTC	N/A	Disabled
1956	1636	ctfmon.exe	0x2231da0	1	86	0	False	2017-05-12 21:22:14.000000 UTC	N/A	Disabled
4	0	System	0x23c8830	51	244	N/A	False	N/A	N/A	Disabled

- **Process Tree**

Displays processes in parent-child tree format to spot suspicious relationships like Word spawning CMD. Raw memory scan found multiple instances of @WanaDecryptor@ at different memory offsets, some already terminated. taskdl.exe and taskse.exe (WannaCry helper processes) were also discovered as hidden/terminated processes.

```
(root@kali) ~/home/kali/Desktop/Practical Investigation Memory Files_1615939758250 (1)/Practical Investigation Memory Files_1615939758250 (1)
python3 /home/kali/volatility3/vol.py -f Investigation-2.raw windows.pstree
```

Volatility 3 Framework 2.28.1
Progress: 100.00 PDB scanning finished

PID	PPID	ImageFileName	Offset(V)	Threads	Handles	SessionId	Wow64	CreateTime	ExitTime	Audit	Cmd	Path
4	0	System	0x823c8830	51	244	N/A	False	N/A	-	-	-	
* 348	4	smss.exe	0x82169020	3	19	N/A	False	2017-05-12 21:21:55.000000 UTC	N/A			\Device\HarddiskVolume1\WINDOWS\system32\smss.exe
tem32\smss.exe		\SystemRoot\System32\smss.exe				\SystemRoot\System32\smss.exe						
** 620	348	winlogon.exe	0x8216e020	23	536	0	False	2017-05-12 21:22:01.000000 UTC	N/A			\Device\HarddiskVolume1\WINDOWS\system32\winlogon.exe
tem32\winlogon.exe		winlogon.exe				\\?\C:\WINDOWS\system32\winlogon.exe						
*** 664	620	services.exe	0x821937f0	15	265	0	False	2017-05-12 21:22:01.000000 UTC	N/A			\Device\HarddiskVolume1\WINDOWS\system32\services.exe
tem32\services.exe		C:\WINDOWS\system32\services.exe				C:\WINDOWS\system32\services.exe						
**** 1024	664	svchost.exe	0x821af7e8	79	1366	0	False	2017-05-12 21:22:03.000000 UTC	N/A			\Device\HarddiskVolume1\WINDOWS\system32\svchost.exe
DOWS\system32\svchost.exe		C:\WINDOWS\system32\svchost.exe				-k netsvcs						C:\WINDOWS\system32\svchost.exe
***** 1768	1024	wuaucit.exe	0x81f747c0	7	132	0	False	2017-05-12 21:22:52.000000 UTC	N/A			\Device\HarddiskVolume1\WINDOWS\system32\wuaucit.exe
DOWS\system32\wuaucit.exe		C:\WINDOWS\system32\wuaucit.exe				/RunStoreAsComServer Local\{400}SUSDS81a6658cb72fa845814e75cca9a42bf2						C:\WINDOWS\system32\wuaucit.exe
***** 1168	1024	wscntfy.exe	0x81fea8a0	1	37	0	False	2017-05-12 21:22:56.000000 UTC	N/A			\Device\HarddiskVolume1\WINDOWS\system32\wscntfy.exe
DOWS\system32\wscntfy.exe		C:\WINDOWS\system32\wscntfy.exe				C:\WINDOWS\system32\wscntfy.exe						
***** 1152	664	svchost.exe	0x821bea78	10	173	0	False	2017-05-12 21:22:06.000000 UTC	N/A			\Device\HarddiskVolume1\WINDOWS\system32\svchost.exe
DOWS\system32\svchost.exe		C:\WINDOWS\system32\svchost.exe				-k LocalService						C:\WINDOWS\system32\svchost.exe
***** 544	664	alg.exe	0x82010020	6	101	0	False	2017-05-12 21:22:55.000000 UTC	N/A			\Device\HarddiskVolume1\WINDOWS\system32\alg.exe
tem32\alg.exe		C:\WINDOWS\system32\alg.exe				C:\WINDOWS\system32\alg.exe						
***** 836	664	svchost.exe	0x8221a2c0	19	211	0	False	2017-05-12 21:22:02.000000 UTC	N/A			\Device\HarddiskVolume1\WINDOWS\system32\svchost.exe
DOWS\system32\svchost.exe		C:\WINDOWS\system32\svchost.exe				-k DcomLaunch						C:\WINDOWS\system32\svchost.exe
***** 260	664	svchost.exe	0x81fb95d8	5	105	0	False	2017-05-12 21:22:18.000000 UTC	N/A			\Device\HarddiskVolume1\WINDOWS\system32\svchost.exe
DOWS\system32\svchost.exe		C:\WINDOWS\system32\svchost.exe				-k LocalService						C:\WINDOWS\system32\svchost.exe
***** 904	664	svchost.exe	0x821b5230	9	227	0	False	2017-05-12 21:22:03.000000 UTC	N/A			\Device\HarddiskVolume1\WINDOWS\system32\svchost.exe
DOWS\system32\svchost.exe		C:\WINDOWS\system32\svchost.exe				-k rpcss						C:\WINDOWS\system32\svchost.exe
***** 1484	664	spoolsv.exe	0x821e2da0	14	124	0	False	2017-05-12 21:22:09.000000 UTC	N/A			\Device\HarddiskVolume1\WINDOWS\system32\spoolsv.exe
DOWS\system32\spoolsv.exe		C:\WINDOWS\system32\spoolsv.exe				C:\WINDOWS\system32\spoolsv.exe						
***** 1084	664	svchost.exe	0x8203b7a8	6	72	0	False	2017-05-12 21:22:03.000000 UTC	N/A			\Device\HarddiskVolume1\WINDOWS\system32\svchost.exe
DOWS\system32\svchost.exe		C:\WINDOWS\system32\svchost.exe				-k NetworkService						C:\WINDOWS\system32\svchost.exe
***** 676	620	lsass.exe	0x2191658	23	353	0	False	2017-05-12 21:22:01.000000 UTC	N/A			\Device\HarddiskVolume1\WINDOWS\system32\lsass.exe
tem32\lsass.exe		C:\WINDOWS\system32\lsass.exe				C:\WINDOWS\system32\lsass.exe						
** 596	348	csrss.exe	0x82161da0	12	352	0	False	2017-05-12 21:22:00.000000 UTC	N/A			\Device\HarddiskVolume1\WINDOWS\system32\csrss.exe
tem32\csrss.exe		C:\WINDOWS\system32\csrss.exe				ObjectDirectory=Windows SharedSection=1024,3072,512 Windows=On SubSystemType=Windows ServerDll=baserv,1 ServerDll=winssrv:UserServerDllInitialization,3 ServerDll=winssrv:ConServerDllInitialization,2 ProfileControl=Off MaxRequestThreads=16						\\?\C:\WINDOWS\system32\csrss.exe
1636	1608	explorer.exe	0x821d9da0	11	331	0	False	2017-05-12 21:22:10.000000 UTC	N/A			\Device\HarddiskVolume1\WINDOWS\explorer.exe
lorer.exe		C:\WINDOWS\Explorer.EXE				C:\WINDOWS\Explorer.EXE						

```

lorer.exe C:\WINDOWS\Explorer.EXE C:\WINDOWS\Explorer.EXE
* 1956 1636 ctfmon.exe 0x82231da0 1 86 0 False 2017-05-12 21:22:14.000000 UTC N/A \Device\HarddiskVolume1\WINDOWS\sys
tem32\ctfmon.exe "C:\WINDOWS\system32\ctfmon.exe" C:\WINDOWS\system32\ctfmon.exe
* 1940 1636 tasksche.exe 0x82218da0 7 51 0 False 2017-05-12 21:22:14.000000 UTC N/A \Device\HarddiskVolume1\Intel\ivecu
qmanpnirkt615\tasksche.exe "C:\Intel\ivecuqmanpnirkt615\tasksche.exe" C:\Intel\ivecuqmanpnirkt615\tasksche.exe
** 740 1940 @WanaDecryptor@ 0x81fde308 2 70 0 False 2017-05-12 21:22:22.000000 UTC N/A \Device\HarddiskVolume1\Intel\ivecu
qmanpnirkt615\@WanaDecryptor@.exe @WanaDecryptor@.exe C:\Intel\ivecuqmanpnirkt615\@WanaDecryptor@.exe

```

- **Command Line Arguments**

Shows full command line used to launch each process revealing malicious scripts or suspicious flags passed to processes. Command line arguments for all processes were extracted.

@WanaDecryptor@ ran with no arguments, while wuaucit.exe was launched with a suspicious RunStoreAsComServer parameter indicating malware impersonation of a Windows update service.

```

(root@kali) - [~/home/kali/Desktop/Practical Investigation Memory Files_1615939758250 (1)/Practical Investigation Memory Files_1615939758250 (1)]
python3 /home/kali/volatility3/vol.py -f Investigation-2.raw windows.cmdline
Volatility 3 Framework 2.28.1
Progress: 100.00 PDB scanning finished
PID Process Args
4 System -
348 smss.exe \SystemRoot\System32\smss.exe
596 csrss.exe C:\WINDOWS\system32\csrss.exe ObjectDirectory=\Windows SharedSection=1024,3072,512 Windows=On SubSystemType=Windows ServerDll=bases
rv,1 ServerDll=winsrv:UserServerDllInitialization,3 ServerDll=winsrv:ConServerDllInitialization,2 ProfileControl=Off MaxRequestThreads=16
620 winlogon.exe winlogon.exe
664 services.exe C:\WINDOWS\system32\services.exe
676 lsass.exe C:\WINDOWS\system32\lsass.exe
836 svchost.exe C:\WINDOWS\system32\svchost -k DcomLaunch
904 svchost.exe C:\WINDOWS\system32\svchost -k rpcss
1024 svchost.exe C:\WINDOWS\system32\svchost.exe -k netsvcs
1084 svchost.exe C:\WINDOWS\system32\svchost.exe -k NetworkService
1152 svchost.exe C:\WINDOWS\system32\svchost.exe -k LocalService
1484 spoolsv.exe C:\WINDOWS\system32\spoolsv.exe
1636 explorer.exe C:\WINDOWS\Explorer.EXE
1940 tasksche.exe "C:\Intel\ivecuqmanpnirkt615\tasksche.exe"
1956 ctfmon.exe "C:\WINDOWS\system32\ctfmon.exe"
260 svchost.exe C:\WINDOWS\system32\svchost.exe -k LocalService
740 @WanaDecryptor@ @WanaDecryptor@.exe
1768 wuaucit.exe "C:\WINDOWS\system32\wuaucit.exe" /RunStoreAsComServer Local\{400}SUSDS81a6658cb72fa845814e75cca9a42bf2
544 alg.exe C:\WINDOWS\System32\alg.exe
1168 wscntfy.exe C:\WINDOWS\system32\wscntfy.exe

```

- **Malware Detection**

Detects injected shellcode or suspicious executable code hidden inside legitimate process memory regions.

Malfind detected suspicious PAGE_EXECUTE_READWRITE memory in csrss.exe (PID 596) and winlogon.exe (PID 620). This indicates code injection into critical Windows system processes, a common malware technique.

```
(root@kali) ~/home/kali/Desktop/Practical Investigation Memory Files_1615939758250 (1)/Practical Investigation Memory Files_1615939758250 (1)
python3 /home/kali/volatility3/vol.py -f Investigation-2.raw windows.malfind
Volatility 3 Framework 2.28.1
/home/kali/volatility3/volatility3/framework/deprecation.py:28: FutureWarning: This API (volatility3.plugins.windows.malfind.Malfind.run) will be removed in the first release after 2026-06-07. This plugin has been renamed, please call volatility3.plugins.windows.malware.malfind.Malfind rather than volatility3.plugins.windows.malfind.Malfind.
warnings.warn()

PID    Process Start VPN    End VPN Tag    Protection    CommitCharge    PrivateMemory    File output    Notes    Hexdump Disasm
/home/kali/volatility3/volatility3/framework/deprecation.py:105: FutureWarning: This plugin (volatility3.plugins.windows.malfind.Malfind) has been renamed and will be removed in the first release after 2026-06-07. Please ensure all method calls to this plugin are replaced with calls to volatility3.plugins.windows.malware.malfind.Malfind
warnings.warn()

596    csrss.exe    0x7f6f0000    0x7f7effff    Vad    PAGE_EXECUTE_READWRITE    0    0    Disabled    N/A
c8 00 00 00 8b 01 00 00 ff ee ff ee 08 70 00 00 .....P..
08 00 00 00 00 fe 00 00 00 00 10 00 00 20 00 00 .....
00 02 00 00 00 20 00 00 8d 01 00 00 ff ef fd 7f .....
03 00 08 06 00 00 00 00 00 00 00 00 00 00 00 00 .....
0x7f6f0000:    enter    0, 0
0x7f6f0004:    mov     eax, dword ptr [ecx]
0x7f6f0006:    add     byte ptr [eax], al
620    winlogon.exe    0x590000    0x5a5ffff    Vad    PAGE_EXECUTE_READ    0    0    Disabled    N/A
c1 00 00 00 00 01 00 00 ff ee ff ee 09 00 00 00 .....
09 00 00 00 00 fe 00 00 00 00 10 00 00 20 00 00 .....
00 02 00 00 00 20 00 00 f3 07 00 00 ff ef fd 7f .....
00 00 08 06 00 00 00 00 00 00 00 00 00 00 00 00 .....
0x590000:    rol     dword ptr [eax], 0
0x590003:    rdd     byte ptr [eax], al
0x590005:    add     dword ptr [eax], eax
0x590007:    add     bh, bh
0x590009:    out     dx, al
620    winlogon.exe    0x21400000    0x21403fff    VadS    PAGE_EXECUTE_READWRITE    4    1    Disabled    N/A
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00 00 00 00 28 00 28 00 01 00 00 00 00 00 00 00 ....(.
0x21400000:    add     byte ptr [eax], al
0x21400002:    add     byte ptr [eax], al
```

Continuation showing more injected memory regions in winlogon.exe with PAGE_EXECUTE_READWRITE protection. The repeated null-byte patterns suggest shellcode or injected payload hiding in legitimate process memory.

```
0x7f6f0000:    enter    0, 0
0x7f6f0004:    mov     eax, dword ptr [ecx]
0x7f6f0006:    add     byte ptr [eax], al
620    winlogon.exe    0x7f630000    0x7f633fff    VadS    PAGE_EXECUTE_READWRITE    4    1    Disabled    N/A
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00 00 00 00 23 00 23 00 01 00 00 00 00 00 00 00 ....##.
0x7f630000:    add     byte ptr [eax], al
0x7f630002:    add     byte ptr [eax], al
0x7f630004:    add     byte ptr [eax], al
0x7f630006:    add     byte ptr [eax], al
0x7f630008:    add     byte ptr [eax], al
0x7f63000a:    add     byte ptr [eax], al
0x7f63000c:    add     byte ptr [eax], al
0x7f63000e:    add     byte ptr [eax], al
0x7f630010:    add     byte ptr [eax], al
0x7f630012:    add     byte ptr [eax], al
0x7f630014:    add     byte ptr [eax], al
0x7f630016:    add     byte ptr [eax], al
0x7f630018:    add     byte ptr [eax], al
0x7f63001a:    add     byte ptr [eax], al
0x7f63001c:    add     byte ptr [eax], al
0x7f63001e:    add     byte ptr [eax], al
0x7f630020:    add     byte ptr [eax], al
0x7f630022:    add     byte ptr [eax], al
0x7f630024:    add     byte ptr [eax], al
0x7f630026:    add     byte ptr [eax], al
0x7f630028:    add     byte ptr [eax], al
0x7f63002a:    add     byte ptr [eax], al
0x7f63002c:    add     byte ptr [eax], al
0x7f63002e:    add     byte ptr [eax], al
0x7f630030:    add     byte ptr [eax], al
0x7f630032:    add     byte ptr [eax], al
0x7f630034:    and     eax, dword ptr [eax]
0x7f630036:    and     eax, dword ptr [eax]
0x7f630038:    add     dword ptr [eax], eax
0x7f63003a:    add     byte ptr [eax], al
0x7f63003c:    add     byte ptr [eax], al
0x7f63003e:    add     byte ptr [eax], al

(root@kali) ~/home/kali/Desktop/Practical Investigation Memory Files_1615939758250 (1)/Practical Investigation Memory Files_1615939758250 (1)
```

- **DLL List**

Lists all DLLs loaded by each process to spot rogue or misspelled DLLs injected by malware. DLL list for smss.exe and csrss.exe shows standard Windows system DLLs like ntdll.dll, KERNEL32.dll, and CSRSRV.dll. These appear normal with expected system32 paths.

```
(root@kali) - [~/home/kali/Desktop/Practical Investigation Memory Files_1615939758250 (1)/Practical Investigation Memory Files_1615939758250 (1)]
# python3 /home/kali/volatility3/vol.py -f Investigation-2.raw windows.dllexport
Volatility 3 Framework 2.28.1
Progress: 100.00
PDB scanning finished
```

PID	Process	Base	Size	Name	Path	LoadCount	LoadTime	File output
348	smss.exe	0x48580000	0xf000	smss.exe	\SystemRoot\System32\smss.exe	-1	N/A	Disabled
348	smss.exe	0x7c900000	0xb2000	ntdll.dll	C:\WINDOWS\system32\ntdll.dll	-1	N/A	Disabled
596	csrss.exe	0x4a680000	0x5000	csrss.exe	\\?\C:\WINDOWS\system32\csrss.exe	-1	N/A	Disabled
596	csrss.exe	0x7c900000	0xb2000	ntdll.dll	C:\WINDOWS\system32\ntdll.dll	-1	N/A	Disabled
596	csrss.exe	0x75b40000	0xb000	CSRSRV.dll	C:\WINDOWS\system32\CSRSRV.dll	-1	N/A	Disabled
596	csrss.exe	0x75b50000	0x10000	basesrv.dll	C:\WINDOWS\system32\basesrv.dll	3	N/A	Disabled
596	csrss.exe	0x75b60000	0x4b000	winsrv.dll	C:\WINDOWS\system32\winsrv.dll	2	N/A	Disabled
596	csrss.exe	0x77f10000	0x49000	GDI32.dll	C:\WINDOWS\system32\GDI32.dll	10	N/A	Disabled
596	csrss.exe	0x7c800000	0xf6000	KERNEL32.dll	C:\WINDOWS\system32\KERNEL32.dll	25	N/A	Disabled
596	csrss.exe	0x7e410000	0x91000	USER32.dll	C:\WINDOWS\system32\USER32.dll	10	N/A	Disabled
596	csrss.exe	0x629c0000	0x9000	LPK.DLL	C:\WINDOWS\system32\LPK.DLL	1	N/A	Disabled
596	csrss.exe	0x74d90000	0x6b000	USP10.dll	C:\WINDOWS\system32\USP10.dll	1	N/A	Disabled
596	csrss.exe	0x77d60000	0x9b000	ADVAPI32.dll	C:\WINDOWS\system32\ADVAPI32.dll	8	N/A	Disabled
596	csrss.exe	0x77e70000	0x93000	RPCRT4.dll	C:\WINDOWS\system32\RPCRT4.dll	4	N/A	Disabled
596	csrss.exe	0x77fe0000	0x11000	Secur32.dll	C:\WINDOWS\system32\Secur32.dll	3	N/A	Disabled
596	csrss.exe	0x7e720000	0xb0000	sxs.dll	C:\WINDOWS\system32\sxs.dll	1	N/A	Disabled
620	winlogon.exe	0x1000000	0x81000	winlogon.exe	\\?\C:\WINDOWS\system32\winlogon.exe	-1	N/A	Disabled
620	winlogon.exe	0x7c900000	0xb2000	ntdll.dll	C:\WINDOWS\system32\ntdll.dll	-1	N/A	Disabled
620	winlogon.exe	0x7c800000	0xf6000	kernel32.dll	C:\WINDOWS\system32\kernel32.dll	-1	N/A	Disabled
620	winlogon.exe	0x77d60000	0x9b000	ADVAPI32.dll	C:\WINDOWS\system32\ADVAPI32.dll	-1	N/A	Disabled
620	winlogon.exe	0x77e70000	0x93000	RPCRT4.dll	C:\WINDOWS\system32\RPCRT4.dll	-1	N/A	Disabled
620	winlogon.exe	0x77fe0000	0x11000	Secur32.dll	C:\WINDOWS\system32\Secur32.dll	-1	N/A	Disabled
620	winlogon.exe	0x776c0000	0x12000	AUTHZ.dll	C:\WINDOWS\system32\AUTHZ.dll	-1	N/A	Disabled
620	winlogon.exe	0x77c10000	0x58000	msvcrt.dll	C:\WINDOWS\system32\msvcrt.dll	-1	N/A	Disabled
620	winlogon.exe	0x77a80000	0x97000	CRYPT32.dll	C:\WINDOWS\system32\CRYPT32.dll	-1	N/A	Disabled
620	winlogon.exe	0x77b20000	0x12000	MSASN1.dll	C:\WINDOWS\system32\MSASN1.dll	-1	N/A	Disabled
620	winlogon.exe	0x7e410000	0x91000	USER32.dll	C:\WINDOWS\system32\USER32.dll	-1	N/A	Disabled
620	winlogon.exe	0x77f10000	0x49000	GDI32.dll	C:\WINDOWS\system32\GDI32.dll	-1	N/A	Disabled
620	winlogon.exe	0x75940000	0x8000	NDdeApi.dll	C:\WINDOWS\system32\NDdeApi.dll	-1	N/A	Disabled
620	winlogon.exe	0x5b860000	0xa000	PROFMAP.dll	C:\WINDOWS\system32\PROFMAP.dll	-1	N/A	Disabled
620	winlogon.exe	0x769c0000	0x56000	NETAPI32.dll	C:\WINDOWS\system32\NETAPI32.dll	-1	N/A	Disabled
620	winlogon.exe	0x76b70000	0xb4000	USERENV.dll	C:\WINDOWS\system32\USERENV.dll	-1	N/A	Disabled
620	winlogon.exe	0x76b70000	0xb000	PSAPI.DLL	C:\WINDOWS\system32\PSAPI.DLL	-1	N/A	Disabled
620	winlogon.exe	0x76b70000	0xf000	REGAPI.dll	C:\WINDOWS\system32\REGAPI.dll	-1	N/A	Disabled
620	winlogon.exe	0x77920000	0xf3000	SETUPAPI.dll	C:\WINDOWS\system32\SETUPAPI.dll	-1	N/A	Disabled
620	winlogon.exe	0x77c00000	0x8000	VERSION.dll	C:\WINDOWS\system32\VERSION.dll	-1	N/A	Disabled
620	winlogon.exe	0x76360000	0x10000	WINSTA.dll	C:\WINDOWS\system32\WINSTA.dll	-1	N/A	Disabled
620	winlogon.exe	0x76c30000	0x2e000	WINTRUST.dll	C:\WINDOWS\system32\WINTRUST.dll	-1	N/A	Disabled
620	winlogon.exe	0x76c90000	0x28000	IMAGEHLP.dll	C:\WINDOWS\system32\IMAGEHLP.dll	-1	N/A	Disabled
620	winlogon.exe	0x71ab0000	0x17000	WS2_32.dll	C:\WINDOWS\system32\WS2_32.dll	-1	N/A	Disabled
620	winlogon.exe	0x71aa0000	0x8000	WS2HELP.dll	C:\WINDOWS\system32\WS2HELP.dll	-1	N/A	Disabled
620	winlogon.exe	0x76390000	0x1d000	IMM32.DLL	C:\WINDOWS\system32\IMM32.DLL	5	N/A	Disabled
620	winlogon.exe	0x629c0000	0x9000	LPK.DLL	C:\WINDOWS\system32\LPK.DLL	1	N/A	Disabled
620	winlogon.exe	0x74d90000	0x6b000	USP10.dll	C:\WINDOWS\system32\USP10.dll	1	N/A	Disabled
620	winlogon.exe	0x75970000	0xf8000	MSGINA.dll	C:\WINDOWS\system32\MSGINA.dll	2	N/A	Disabled
620	winlogon.exe	0x5d090000	0x9a000	COMCTL32.dll	C:\WINDOWS\system32\COMCTL32.dll	7	N/A	Disabled
620	winlogon.exe	0x74220000	0x3e000	ODBC32.dll	C:\WINDOWS\system32\ODBC32.dll	2	N/A	Disabled
620	winlogon.exe	0x763b0000	0x49000	comdlg32.dll	C:\WINDOWS\system32\comdlg32.dll	2	N/A	Disabled
620	winlogon.exe	0x76c90000	0x818000	SHELL32.dll	C:\WINDOWS\system32\SHELL32.dll	16	N/A	Disabled
620	winlogon.exe	0x77f60000	0x76000	SHLWAPI.dll	C:\WINDOWS\system32\SHLWAPI.dll	27	N/A	Disabled
620	winlogon.exe	0x773d0000	0x103000	comctl32.dll	C:\WINDOWS\WinSxS\x86_Microsoft.Windows.Common-Controls_6595b64144ccf1df_6.0.2600.6			
028_x-w_61e65202\comctl32.dll	3	N/A	Disabled					
620	winlogon.exe	0x9a0000	0x17000	odbcint.dll	C:\WINDOWS\system32\odbcint.dll	1	N/A	Disabled
620	winlogon.exe	0x776e0000	0x23000	SHSVCS.dll	C:\WINDOWS\system32\SHSVCS.dll	1	N/A	Disabled
620	winlogon.exe	0x76bb0000	0x5000	sfc.dll	C:\WINDOWS\system32\sfc.dll	2	N/A	Disabled
620	winlogon.exe	0x76c60000	0x2a000	sfc_os.dll	C:\WINDOWS\system32\sfc_os.dll	5	N/A	Disabled
620	winlogon.exe	0x774e0000	0x13e000	ole32.dll	C:\WINDOWS\system32\ole32.dll	21	N/A	Disabled
620	winlogon.exe	0x77b40000	0x22000	Apphelp.dll	C:\WINDOWS\system32\Apphelp.dll	1	N/A	Disabled

services.exe (PID 664) loaded standard service management DLLs. The presence of NCOBJAPI.DLL, SCESRV.dll, and ShimEng.dll all appear legitimate for a Windows XP services process.

664	services.exe	0x1000000	0x1d000	services.exe	C:\WINDOWS\system32\services.exe	-1	N/A	Disabled
664	services.exe	0x7c900000	0xb2000	ntdll.dll	C:\WINDOWS\system32\ntdll.dll	-1	N/A	Disabled
664	services.exe	0x7c800000	0xf6000	kernel32.dll	C:\WINDOWS\system32\kernel32.dll	-1	N/A	Disabled
664	services.exe	0x77dd0000	0x9b000	ADVAPI32.dll	C:\WINDOWS\system32\ADVAPI32.dll	-1	N/A	Disabled
664	services.exe	0x77e70000	0x93000	RPCRT4.dll	C:\WINDOWS\system32\RPCRT4.dll	-1	N/A	Disabled
664	services.exe	0x77fe0000	0x11000	Secur32.dll	C:\WINDOWS\system32\Secur32.dll	-1	N/A	Disabled
664	services.exe	0x77f10000	0x58000	msvcrt.dll	C:\WINDOWS\system32\msvcrt.dll	-1	N/A	Disabled
664	services.exe	0x5f770000	0xc000	NCObjAPI.DLL	C:\WINDOWS\system32\NCObjAPI.DLL	-1	N/A	Disabled
664	services.exe	0x76080000	0x65000	MSVCP60.dll	C:\WINDOWS\system32\MSVCP60.dll	-1	N/A	Disabled
664	services.exe	0x7dbd0000	0x51000	SCESRV.dll	C:\WINDOWS\system32\SCESRV.dll	-1	N/A	Disabled
664	services.exe	0x776c0000	0x12000	AUTHZ.dll	C:\WINDOWS\system32\AUTHZ.dll	-1	N/A	Disabled
664	services.exe	0x7e410000	0x91000	USER32.dll	C:\WINDOWS\system32\USER32.dll	-1	N/A	Disabled
664	services.exe	0x77f10000	0x49000	GDI32.dll	C:\WINDOWS\system32\GDI32.dll	-1	N/A	Disabled
664	services.exe	0x769c0000	0xb4000	USERENV.dll	C:\WINDOWS\system32\USERENV.dll	-1	N/A	Disabled
664	services.exe	0x7dba0000	0x21000	umpnpmgr.dll	C:\WINDOWS\system32\umpnpmgr.dll	-1	N/A	Disabled
664	services.exe	0x76b60000	0x10000	WINSTA.dll	C:\WINDOWS\system32\WINSTA.dll	-1	N/A	Disabled
664	services.exe	0x5b860000	0x56000	NETAPI32.dll	C:\WINDOWS\system32\NETAPI32.dll	-1	N/A	Disabled
664	services.exe	0x5cb70000	0x26000	ShimEng.dll	C:\WINDOWS\system32\ShimEng.dll	1	N/A	Disabled
664	services.exe	0x47260000	0xf000	AcAdProc.dll	C:\WINDOWS\AppPatch\AcAdProc.dll	1	N/A	Disabled
664	services.exe	0x76390000	0x1d000	IMM32.DLL	C:\WINDOWS\system32\IMM32.DLL	1	N/A	Disabled
664	services.exe	0x629c0000	0x9000	LPK.DLL	C:\WINDOWS\system32\LPK.DLL	1	N/A	Disabled
664	services.exe	0x74d90000	0x6b000	USP10.dll	C:\WINDOWS\system32\USP10.dll	1	N/A	Disabled
664	services.exe	0x59c00000	0x7000	credssp.dll	C:\WINDOWS\system32\credssp.dll	1	N/A	Disabled
664	services.exe	0x77a80000	0x97000	CRYPT32.dll	C:\WINDOWS\system32\CRYPT32.dll	2	N/A	Disabled
664	services.exe	0x77b20000	0x12000	MSASN1.dll	C:\WINDOWS\system32\MSASN1.dll	3	N/A	Disabled
664	services.exe	0x767f0000	0x29000	schannel.dll	C:\WINDOWS\system32\schannel.dll	1	N/A	Disabled
664	services.exe	0x71ab0000	0x17000	WS2_32.dll	C:\WINDOWS\system32\WS2_32.dll	2	N/A	Disabled
664	services.exe	0x71aa0000	0x8000	WS2HELP.dll	C:\WINDOWS\system32\WS2HELP.dll	2	N/A	Disabled
664	services.exe	0x77ba0000	0x22000	Apphelp.dll	C:\WINDOWS\system32\Apphelp.dll	1	N/A	Disabled

lsass.exe (PID 676) shows credential-related DLLs like kerberos.dll, msv1_0.dll, credssp.dll, and wdigest.dll. These are normal for the Local Security Authority process handling authentication.

676	lsass.exe	0x77be0000	0x15000	MSACM32.dll	C:\WINDOWS\system32\MSACM32.dll	1	N/A	Disabled
676	lsass.exe	0x77c00000	0x8000	VERSION.dll	C:\WINDOWS\system32\VERSION.dll	1	N/A	Disabled
676	lsass.exe	0x7c9c0000	0x818000	SHELL32.dll	C:\WINDOWS\system32\SHELL32.dll	2	N/A	Disabled
676	lsass.exe	0x77f60000	0x76000	SHLWAPI.dll	C:\WINDOWS\system32\SHLWAPI.dll	4	N/A	Disabled
676	lsass.exe	0x769c0000	0xb4000	USERENV.dll	C:\WINDOWS\system32\USERENV.dll	15	N/A	Disabled
676	lsass.exe	0x5ad70000	0x38000	UxTheme.dll	C:\WINDOWS\system32\UxTheme.dll	3	N/A	Disabled
676	lsass.exe	0x76390000	0x1d000	IMM32.DLL	C:\WINDOWS\system32\IMM32.DLL	2	N/A	Disabled
676	lsass.exe	0x629c0000	0x9000	LPK.DLL	C:\WINDOWS\system32\LPK.DLL	1	N/A	Disabled
676	lsass.exe	0x74d90000	0x6b000	USP10.dll	C:\WINDOWS\system32\USP10.dll	1	N/A	Disabled
676	lsass.exe	0x773d0000	0x103000	comctl32.dll	C:\WINDOWS\WinSxS\x86_Microsoft.Windows.Common-Controls_6595b64144ccf1df_6.0.2600.6_028_x-ww_61e65202\comctl32.dll	1	N/A	Disabled
676	lsass.exe	0x5d090000	0x9a000	comctl32.dll	C:\WINDOWS\system32\comctl32.dll	1	N/A	Disabled
676	lsass.exe	0x59c00000	0x7000	credssp.dll	C:\WINDOWS\system32\credssp.dll	2	N/A	Disabled
676	lsass.exe	0x77a80000	0x97000	CRYPT32.dll	C:\WINDOWS\system32\CRYPT32.dll	11	N/A	Disabled
676	lsass.exe	0x71e50000	0x15000	msapsspc.dll	C:\WINDOWS\system32\msapsspc.dll	1	N/A	Disabled
676	lsass.exe	0x78080000	0x11000	MSVCRT40.dll	C:\WINDOWS\system32\MSVCRT40.dll	2	N/A	Disabled
676	lsass.exe	0x75b00000	0x47000	digest.dll	C:\WINDOWS\system32\digest.dll	1	N/A	Disabled
676	lsass.exe	0x747b0000	0x47000	msnsspc.dll	C:\WINDOWS\system32\msnsspc.dll	1	N/A	Disabled
676	lsass.exe	0x4d200000	0xe000	msprivs.dll	C:\WINDOWS\system32\msprivs.dll	1	N/A	Disabled
676	lsass.exe	0x77c70000	0x4c000	kerberos.dll	C:\WINDOWS\system32\kerberos.dll	2	N/A	Disabled
676	lsass.exe	0x77f10000	0x25000	msv1_0.dll	C:\WINDOWS\system32\msv1_0.dll	5	N/A	Disabled
676	lsass.exe	0x76d60000	0x19000	iphlpapi.dll	C:\WINDOWS\system32\iphlpapi.dll	8	N/A	Disabled
676	lsass.exe	0x744b0000	0x65000	netlogon.dll	C:\WINDOWS\system32\netlogon.dll	2	N/A	Disabled
676	lsass.exe	0x767c0000	0x2d000	w32time.dll	C:\WINDOWS\system32\w32time.dll	2	N/A	Disabled
676	lsass.exe	0x76080000	0x65000	MSVCP60.dll	C:\WINDOWS\system32\MSVCP60.dll	2	N/A	Disabled
676	lsass.exe	0x767f0000	0x29000	schannel.dll	C:\WINDOWS\system32\schannel.dll	7	N/A	Disabled
676	lsass.exe	0x74df0000	0x11000	wdigest.dll	C:\WINDOWS\system32\wdigest.dll	1	N/A	Disabled
676	lsass.exe	0x68000000	0x36000	rsaenh.dll	C:\WINDOWS\system32\rsaenh.dll	1	N/A	Disabled
676	lsass.exe	0x723d0000	0xf000	tspkg.dll	C:\WINDOWS\system32\tspkg.dll	1	N/A	Disabled
676	lsass.exe	0x723d0000	0x1c000	WinSCard.dll	C:\WINDOWS\system32\WinSCard.dll	1	N/A	Disabled
676	lsass.exe	0x76f50000	0x8000	WTSAPI32.dll	C:\WINDOWS\system32\WTSAPI32.dll	1	N/A	Disabled
676	lsass.exe	0x76360000	0x10000	WINSTA.dll	C:\WINDOWS\system32\WINSTA.dll	1	N/A	Disabled
676	lsass.exe	0x74410000	0x2f000	scecli.dll	C:\WINDOWS\system32\scecli.dll	1	N/A	Disabled
676	lsass.exe	0x77920000	0xf3000	SETUPAPI.dll	C:\WINDOWS\system32\SETUPAPI.dll	1	N/A	Disabled
676	lsass.exe	0x743e0000	0x2f000	ipsecsvc.dll	C:\WINDOWS\system32\ipsecsvc.dll	1	N/A	Disabled

DLLs for multiple svchost.exe instances are listed, all loading from legitimate C:\WINDOWS\system32 paths. No suspicious or out-of-place DLLs detected in these service host processes.

836	svchost.exe	0x77a80000	0x97000	CRYPT32.dll	C:\WINDOWS\system32\CRYPT32.dll	5	N/A	Disabled
836	svchost.exe	0x77b20000	0x12000	MSASN1.dll	C:\WINDOWS\system32\MSASN1.dll	5	N/A	Disabled
836	svchost.exe	0x767f0000	0x29000	schannel.dll	C:\WINDOWS\system32\schannel.dll	1	N/A	Disabled
836	svchost.exe	0x5b860000	0x56000	NETAPI32.dll	C:\WINDOWS\system32\NETAPI32.dll	4	N/A	Disabled
836	svchost.exe	0x76fd0000	0x7f000	CLBCATQ.DLL	C:\WINDOWS\system32\CLBCATQ.DLL	2	N/A	Disabled
836	svchost.exe	0x77050000	0xc5000	COMRes.dll	C:\WINDOWS\system32\COMRes.dll	2	N/A	Disabled
836	svchost.exe	0x760f0000	0x53000	termsrv.dll	c:\windows\system32\termsrv.dll	1	N/A	Disabled
836	svchost.exe	0x74f70000	0x6000	ICAAPI.dll	c:\windows\system32\ICAAPI.dll	1	N/A	Disabled
836	svchost.exe	0x77920000	0xf3000	SETUPAPI.dll	c:\windows\system32\SETUPAPI.dll	1	N/A	Disabled
836	svchost.exe	0x76c30000	0x2e000	WINTRUST.dll	c:\windows\system32\WINTRUST.dll	1	N/A	Disabled
836	svchost.exe	0x76c90000	0x28000	IMAGEHLP.dll	C:\WINDOWS\system32\IMAGEHLP.dll	1	N/A	Disabled
836	svchost.exe	0x77760000	0x12000	AUTHZ.dll	c:\windows\system32\AUTHZ.dll	1	N/A	Disabled
836	svchost.exe	0x75110000	0x1f000	mstlsapi.dll	c:\windows\system32\mstlsapi.dll	1	N/A	Disabled
836	svchost.exe	0x777c0000	0x32000	ACTIVEDS.dll	c:\windows\system32\ACTIVEDS.dll	1	N/A	Disabled
836	svchost.exe	0x76e10000	0x25000	adslrpc.dll	c:\windows\system32\adslrpc.dll	1	N/A	Disabled
836	svchost.exe	0x76b20000	0x11000	ATL.DLL	c:\windows\system32\ATL.DLL	1	N/A	Disabled
836	svchost.exe	0x76bc0000	0xf000	REGAPI.dll	C:\WINDOWS\system32\REGAPI.dll	1	N/A	Disabled
836	svchost.exe	0x68000000	0x36000	rsaenh.dll	C:\WINDOWS\system32\rsaenh.dll	1	N/A	Disabled
836	svchost.exe	0x77ba0000	0x22000	Apphelp.dll	C:\WINDOWS\system32\Apphelp.dll	1	N/A	Disabled
904	svchost.exe	0x10000000	0x6000	svchost.exe	C:\WINDOWS\system32\svchost.exe	-1	N/A	Disabled
904	svchost.exe	0x7c900000	0xb2000	ntdll.dll	C:\WINDOWS\system32\ntdll.dll	-1	N/A	Disabled
904	svchost.exe	0x7c800000	0xf6000	kernel32.dll	C:\WINDOWS\system32\kernel32.dll	-1	N/A	Disabled
904	svchost.exe	0x77dd0000	0x9b000	ADVAPI32.dll	C:\WINDOWS\system32\ADVAPI32.dll	-1	N/A	Disabled
904	svchost.exe	0x777e0000	0x93000	RPCRT4.dll	C:\WINDOWS\system32\RPCRT4.dll	-1	N/A	Disabled
904	svchost.exe	0x77fe0000	0x11000	Secur32.dll	C:\WINDOWS\system32\Secur32.dll	-1	N/A	Disabled
904	svchost.exe	0x5cb70000	0x26000	ShimEng.dll	C:\WINDOWS\system32\ShimEng.dll	1	N/A	Disabled
904	svchost.exe	0x6f880000	0x1ca000	AcGenral.DLL	C:\WINDOWS\AppPatch\AcGenral.DLL	1	N/A	Disabled
904	svchost.exe	0x76410000	0x91000	USER32.dll	C:\WINDOWS\system32\USER32.dll	48	N/A	Disabled
904	svchost.exe	0x77f10000	0x49000	GDI32.dll	C:\WINDOWS\system32\GDI32.dll	37	N/A	Disabled
904	svchost.exe	0x76b40000	0x2d000	WINMM.dll	C:\WINDOWS\system32\WINMM.dll	2	N/A	Disabled
904	svchost.exe	0x774e0000	0x13e000	ole32.dll	C:\WINDOWS\system32\ole32.dll	6	N/A	Disabled
904	svchost.exe	0x777c10000	0x58000	msvcrt.dll	C:\WINDOWS\system32\msvcrt.dll	84	N/A	Disabled
904	svchost.exe	0x77120000	0x8b000	OLEAUT32.dll	C:\WINDOWS\system32\OLEAUT32.dll	3	N/A	Disabled
904	svchost.exe	0x777be0000	0x15000	MSACM32.dll	C:\WINDOWS\system32\MSACM32.dll	1	N/A	Disabled
904	svchost.exe	0x77c00000	0x8000	VERSION.dll	C:\WINDOWS\system32\VERSION.dll	3	N/A	Disabled
904	svchost.exe	0x769c0000	0x818000	SHELL32.dll	C:\WINDOWS\system32\SHELL32.dll	1	N/A	Disabled
904	svchost.exe	0x77f60000	0x76000	SHLWAPI.dll	C:\WINDOWS\system32\SHLWAPI.dll	3	N/A	Disabled
904	svchost.exe	0x76990000	0xb4000	USERENV.dll	C:\WINDOWS\system32\USERENV.dll	3	N/A	Disabled
904	svchost.exe	0x5ad70000	0x38000	UxTheme.dll	C:\WINDOWS\system32\UxTheme.dll	1	N/A	Disabled
904	svchost.exe	0x76390000	0x1d000	IMM32.DLL	C:\WINDOWS\system32\IMM32.DLL	2	N/A	Disabled
904	svchost.exe	0x629c0000	0x9000	LPK.DLL	C:\WINDOWS\system32\LPK.DLL	1	N/A	Disabled
904	svchost.exe	0x74d90000	0x6b000	USP10.dll	C:\WINDOWS\system32\USP10.dll	1	N/A	Disabled
904	svchost.exe	0x773d0000	0x103000	comctl32.dll	C:\WINDOWS\WinSxS\x86_Microsoft.Windows.Common-Controls_6595b64144ccf1df_6.0.2600.6			

explorer.exe (PID 1636) loaded standard shell DLLs including ieiframe.dll, shimvw.dll, and gdiplus.dll. These are all expected components of the Windows Explorer shell.

1636	explorer.exe	0x76b40000	0x2d000	WINMM.dll	C:\WINDOWS\system32\WINMM.dll	12	N/A	Disabled
1636	explorer.exe	0x77be0000	0x15000	MSACM32.dll	C:\WINDOWS\system32\MSACM32.dll	3	N/A	Disabled
1636	explorer.exe	0x76990000	0xb4000	USERENV.dll	C:\WINDOWS\system32\USERENV.dll	5	N/A	Disabled
1636	explorer.exe	0x76390000	0x1d000	IMM32.DLL	C:\WINDOWS\system32\IMM32.DLL	4	N/A	Disabled
1636	explorer.exe	0x629c0000	0x9000	LPK.DLL	C:\WINDOWS\system32\LPK.DLL	1	N/A	Disabled
1636	explorer.exe	0x74d90000	0x6b000	USP10.dll	C:\WINDOWS\system32\USP10.dll	1	N/A	Disabled
1636	explorer.exe	0x773d0000	0x103000	comctl32.dll	C:\WINDOWS\WinSxS\x86_Microsoft.Windows.Common-Controls_6595b64144ccf1df_6.0.2600.6			
028_x-ww_61e65202\comctl32.dll	22	N/A	Disabled					
1636	explorer.exe	0x5d090000	0x9a000	comctl32.dll	C:\WINDOWS\system32\comctl32.dll	2	N/A	Disabled
1636	explorer.exe	0x755c0000	0x2e000	msctfime.ime	C:\WINDOWS\system32\msctfime.ime	2	N/A	Disabled
1636	explorer.exe	0x77b40000	0x22000	apphelp.dll	C:\WINDOWS\system32\apphelp.dll	2	N/A	Disabled
1636	explorer.exe	0x76fd0000	0x7f000	CLBCATQ.DLL	C:\WINDOWS\system32\CLBCATQ.DLL	2	N/A	Disabled
1636	explorer.exe	0x77050000	0xc5000	COMRes.dll	C:\WINDOWS\system32\COMRes.dll	2	N/A	Disabled
1636	explorer.exe	0x77a20000	0x54000	cscui.dll	C:\WINDOWS\system32\cscui.dll	2	N/A	Disabled
1636	explorer.exe	0x76600000	0x1d000	CSCDLL.dll	C:\WINDOWS\system32\CSCDLL.dll	2	N/A	Disabled
1636	explorer.exe	0x5ba60000	0x71000	themeui.dll	C:\WINDOWS\system32\themeui.dll	1	N/A	Disabled
1636	explorer.exe	0x76380000	0x5000	MSIMG32.dll	C:\WINDOWS\system32\MSIMG32.dll	1	N/A	Disabled
1636	explorer.exe	0x11000000	0x2c5000	xpsp2res.dll	C:\WINDOWS\system32\xpsp2res.dll	1	N/A	Disabled
1636	explorer.exe	0x71d40000	0x1b000	ACTXPRXY.DLL	C:\WINDOWS\system32\ACTXPRXY.DLL	1	N/A	Disabled
1636	explorer.exe	0x5fc10000	0x33000	msutb.dll	C:\WINDOWS\system32\msutb.dll	1	N/A	Disabled
1636	explorer.exe	0x74720000	0x4c000	MSCTF.dll	C:\WINDOWS\system32\MSCTF.dll	3	N/A	Disabled
1636	explorer.exe	0x71bf0000	0x13000	SAMLIB.dll	C:\WINDOWS\system32\SAMLIB.dll	2	N/A	Disabled
1636	explorer.exe	0x76990000	0x8000	LINKINFO.dll	C:\WINDOWS\system32\LINKINFO.dll	1	N/A	Disabled
1636	explorer.exe	0x76990000	0x25000	ntshrui.dll	C:\WINDOWS\system32\ntshrui.dll	2	N/A	Disabled
1636	explorer.exe	0x76b20000	0x11000	ATL.DLL	C:\WINDOWS\system32\ATL.DLL	3	N/A	Disabled
1636	explorer.exe	0x3e1c0000	0xa9d000	ieiframe.dll	C:\WINDOWS\system32\ieiframe.dll	3	N/A	Disabled
1636	explorer.exe	0x5cb00000	0x6e000	shimvw.dll	C:\WINDOWS\system32\shimvw.dll	1	N/A	Disabled
1636	explorer.exe	0x4ec50000	0x1ab000	gdiplus.dll	C:\WINDOWS\WinSxS\x86_Microsoft.Windows.GdiPlus_6595b64144ccf1df_1.0.6002.23084_x-w			
w_f3f35550\gdiplus.dll	1	N/A	Disabled					
1636	explorer.exe	0x77920000	0xf3000	SETUPAPI.dll	C:\WINDOWS\system32\SETUPAPI.dll	6	N/A	Disabled
1636	explorer.exe	0x3fde0000	0x440000	msi.dll	C:\WINDOWS\system32\msi.dll	1	N/A	Disabled
1636	explorer.exe	0x76400000	0x1a5000	NETSHELL.dll	C:\WINDOWS\system32\NETSHELL.dll	1	N/A	Disabled

The @WanaDecryptor@ process (PID 740) loaded standard Windows DLLs like MFC42.DLL, OLEAUT32.dll, and WININET.dll. It ran from the suspicious path C:\Intel\ivecuqmanpnirkt615\@WanaDecryptor@.exe.

740	@WanaDecryptor@	0x400000	0x3d000	@WanaDecryptor@.exe	C:\Intel\ivecuqmanpnirkt615\@WanaDecryptor@.exe	-1	N/A	Disabled
740	@WanaDecryptor@	0x7c900000	0xb2000	ntdll.dll	C:\WINDOWS\system32\ntdll.dll	-1	N/A	Disabled
740	@WanaDecryptor@	0x7c800000	0xf6000	kernel32.dll	C:\WINDOWS\system32\kernel32.dll	-1	N/A	Disabled
740	@WanaDecryptor@	0x73dd0000	0xf2000	MFC42.DLL	C:\WINDOWS\system32\MFC42.DLL	-1	N/A	Disabled
740	@WanaDecryptor@	0x77c10000	0x58000	msvcrt.dll	C:\WINDOWS\system32\msvcrt.dll	-1	N/A	Disabled
740	@WanaDecryptor@	0x77f10000	0x49000	GDI32.dll	C:\WINDOWS\system32\GDI32.dll	-1	N/A	Disabled
740	@WanaDecryptor@	0x7e410000	0x91000	USER32.dll	C:\WINDOWS\system32\USER32.dll	-1	N/A	Disabled
740	@WanaDecryptor@	0x77dd0000	0x9b000	ADVAPI32.dll	C:\WINDOWS\system32\ADVAPI32.dll	-1	N/A	Disabled
740	@WanaDecryptor@	0x77e70000	0x93000	RPCRT4.dll	C:\WINDOWS\system32\RPCRT4.dll	-1	N/A	Disabled
740	@WanaDecryptor@	0x77fe0000	0x11000	Secur32.dll	C:\WINDOWS\system32\Secur32.dll	-1	N/A	Disabled
740	@WanaDecryptor@	0x7ec90000	0x818000	SHELL32.dll	C:\WINDOWS\system32\SHELL32.dll	-1	N/A	Disabled
740	@WanaDecryptor@	0x77f60000	0x76000	SHLWAPI.dll	C:\WINDOWS\system32\SHLWAPI.dll	-1	N/A	Disabled
740	@WanaDecryptor@	0x773d0000	0x103000	COMCTL32.dll	C:\WINDOWS\WinSxS\X86_Microsoft.Windows.Common-Controls_6595b64144ccf1df_6.0.2600.6	-1	N/A	Disabled
028_x-ww_61e65202\COMCTL32.dll	-1	N/A	Disabled					
740	@WanaDecryptor@	0x77120000	0x8b000	OLEAUT32.dll	C:\WINDOWS\system32\OLEAUT32.dll	-1	N/A	Disabled
740	@WanaDecryptor@	0x774e0000	0x13e000	ole32.dll	C:\WINDOWS\system32\ole32.dll	-1	N/A	Disabled
740	@WanaDecryptor@	0x78130000	0x134000	urlmon.dll	C:\WINDOWS\system32\urlmon.dll	-1	N/A	Disabled
740	@WanaDecryptor@	0x3dfd0000	0x1ec000	iertutil.dll	C:\WINDOWS\system32\iertutil.dll	-1	N/A	Disabled
740	@WanaDecryptor@	0x76080000	0x65000	MSVCP60.dll	C:\WINDOWS\system32\MSVCP60.dll	-1	N/A	Disabled
740	@WanaDecryptor@	0x71ab0000	0x17000	WS2_32.dll	C:\WINDOWS\system32\WS2_32.dll	-1	N/A	Disabled
740	@WanaDecryptor@	0x71aa0000	0x8000	WS2HELP.dll	C:\WINDOWS\system32\WS2HELP.dll	-1	N/A	Disabled
740	@WanaDecryptor@	0x3d930000	0xe7000	WININET.dll	C:\WINDOWS\system32\WININET.dll	-1	N/A	Disabled
740	@WanaDecryptor@	0x340000	0x9000	Normaliz.dll	C:\WINDOWS\system32\Normaliz.dll	-1	N/A	Disabled
740	@WanaDecryptor@	0x76390000	0x1d000	IMM32.DLL	C:\WINDOWS\system32\IMM32.DLL	4	N/A	Disabled
740	@WanaDecryptor@	0x76290000	0x9000	LPK.DLL	C:\WINDOWS\system32\LPK.DLL	1	N/A	Disabled
740	@WanaDecryptor@	0x74d90000	0x6b000	USP10.dll	C:\WINDOWS\system32\USP10.dll	2	N/A	Disabled
740	@WanaDecryptor@	0x732e0000	0x5000	RICHED32.DLL	C:\WINDOWS\system32\RICHED32.DLL	1	N/A	Disabled
740	@WanaDecryptor@	0x74e30000	0x6d000	RICHED20.dll	C:\WINDOWS\system32\RICHED20.dll	1	N/A	Disabled
740	@WanaDecryptor@	0x5ad70000	0x38000	uxtheme.dll	C:\WINDOWS\system32\uxtheme.dll	3	N/A	Disabled
740	@WanaDecryptor@	0x74720000	0x4c000	MSCTF.dll	C:\WINDOWS\system32\MSCTF.dll	1	N/A	Disabled
740	@WanaDecryptor@	0x755c0000	0x2e000	msctfime.ime	C:\WINDOWS\system32\msctfime.ime	2	N/A	Disabled
740	@WanaDecryptor@	0x769c0000	0xb4000	USERENV.dll	C:\WINDOWS\system32\USERENV.dll	1	N/A	Disabled
740	@WanaDecryptor@	0xea0000	0x29000	msls31.dll	C:\WINDOWS\system32\msls31.dll	1	N/A	Disabled

wscntfy.exe (PID 1168, Windows Security Center notify) loaded normal system DLLs. No anomalies detected in this process's loaded libraries.

1168	wscntfy.exe	0x1000000	0x6000	wscntfy.exe	C:\WINDOWS\system32\wscntfy.exe	-1	N/A	Disabled
1168	wscntfy.exe	0x7c900000	0xb2000	ntdll.dll	C:\WINDOWS\system32\ntdll.dll	-1	N/A	Disabled
1168	wscntfy.exe	0x7c800000	0xf6000	kernel32.dll	C:\WINDOWS\system32\kernel32.dll	-1	N/A	Disabled
1168	wscntfy.exe	0x77c10000	0x58000	msvcrt.dll	C:\WINDOWS\system32\msvcrt.dll	-1	N/A	Disabled
1168	wscntfy.exe	0x7e410000	0x91000	USER32.dll	C:\WINDOWS\system32\USER32.dll	-1	N/A	Disabled
1168	wscntfy.exe	0x77f10000	0x49000	GDI32.dll	C:\WINDOWS\system32\GDI32.dll	-1	N/A	Disabled
1168	wscntfy.exe	0x7ec90000	0x818000	SHELL32.dll	C:\WINDOWS\system32\SHELL32.dll	-1	N/A	Disabled
1168	wscntfy.exe	0x77dd0000	0x9b000	ADVAPI32.dll	C:\WINDOWS\system32\ADVAPI32.dll	-1	N/A	Disabled
1168	wscntfy.exe	0x77e70000	0x93000	RPCRT4.dll	C:\WINDOWS\system32\RPCRT4.dll	-1	N/A	Disabled
1168	wscntfy.exe	0x77fe0000	0x11000	Secur32.dll	C:\WINDOWS\system32\Secur32.dll	-1	N/A	Disabled
1168	wscntfy.exe	0x77f60000	0x76000	SHLWAPI.dll	C:\WINDOWS\system32\SHLWAPI.dll	-1	N/A	Disabled
1168	wscntfy.exe	0x76390000	0x1d000	IMM32.DLL	C:\WINDOWS\system32\IMM32.DLL	3	N/A	Disabled
1168	wscntfy.exe	0x76290000	0x9000	LPK.DLL	C:\WINDOWS\system32\LPK.DLL	1	N/A	Disabled
1168	wscntfy.exe	0x74d90000	0x6b000	USP10.dll	C:\WINDOWS\system32\USP10.dll	1	N/A	Disabled
1168	wscntfy.exe	0x773d0000	0x103000	comctl32.dll	C:\WINDOWS\WinSxS\X86_Microsoft.Windows.Common-Controls_6595b64144ccf1df_6.0.2600.6	-1	N/A	Disabled
028_x-ww_61e65202\comctl32.dll	2	N/A	Disabled					
1168	wscntfy.exe	0x7d0000	0x2c5000	xpsp2res.dll	C:\WINDOWS\system32\xpsp2res.dll	1	N/A	Disabled
1168	wscntfy.exe	0x5ad70000	0x38000	uxtheme.dll	C:\WINDOWS\system32\uxtheme.dll	2	N/A	Disabled
1168	wscntfy.exe	0x74720000	0x4c000	MSCTF.dll	C:\WINDOWS\system32\MSCTF.dll	1	N/A	Disabled
1168	wscntfy.exe	0x755c0000	0x2e000	msctfime.ime	C:\WINDOWS\system32\msctfime.ime	2	N/A	Disabled
1168	wscntfy.exe	0x774e0000	0x13e000	ole32.dll	C:\WINDOWS\system32\ole32.dll	1	N/A	Disabled

- **File Scan**

Scans memory for all file objects that were open at capture time to find dropped malware or suspicious files.

File scan began listing all file objects in memory including system DLLs, prefetch files, and boot.ini. Early offsets show WMI-related files and standard Windows components.

```

(root@kali) ~/home/kali/Desktop/Practical Investigation Memory Files_1615939758250 (1)/Practical Investigation Memory Files_1615939758250 (1)
$ python3 /home/kali/volatility3/vol.py -f Investigation-2.raw windows.filescan
Volatility 3 Framework 2.28.1
Progress: 100.00 PDB scanning finished
Offset Name
0x1f40310 \Endpoint
0x1f65718 \Endpoint
0x1f66cd8 \WINDOWS\system32\wbem\wmipcma.dll
0x1f67198 \WINDOWS\Prefetch\TASKDL.EXE-01687054.pf
0x1f67a70 \WINDOWS\system32\security.dll
0x1f67c68 \boot.ini
0x1f67ef8 \WINDOWS\system32\cfgmgr32.dll
0x1f684d0 \WINDOWS\system32\wbem\framedyn.dll
0x1f686d8 \WINDOWS\system32\wbem\cimwin32.dll
0x1f6a7f0 \WINDOWS\system32\kmdhlp.dll
0x1f6ae20 \Directory
0x1f6b9b0 \Directory
0x1f6bbf8 \Directory
0x1f6bdc8 \PIPE_EVENTROOT\CIMV2SCM_EVENT_PROVIDER
0x1f6be60 \WINDOWS\win.ini

```

Filescan revealed interesting entries including SAM database, user Desktop files, and a suspicious file PIL-1.1.7.win32-py2.7.exe on the user donny's Desktop. The username donny identifies the victim machine's user account.

```

0x239b928 \WINDOWS\system32\config\SAM.LOG
0x239bb80 \WINDOWS\system32\config\SAM
0x239c690 \WINDOWS\system32\sfcdll.dll
0x239c790 \Directory
0x239f478 \WINDOWS\system32\dfgrres.dll
0x239f6d0 \Documents and Settings\donny\Desktop
0x239f928 \Documents and Settings\donny\PrintHood
0x239fb80 \WINDOWS\WinSxS\x86_Microsoft.Windows.Common-Controls_6595b64144ccf1df_6.0.2600.6028_x-ww_61e65202
0x239fc78 \Directory
0x239ff28 \WINDOWS\system32
0x23a0238 \WINDOWS\system32\tourstart.exe
0x23a0370 \WINDOWS\hh.exe
0x23a0788 \Documents and Settings\All Users\Start Menu\Microsoft Update.lnk
0x23a0920 \WINDOWS\system32\msxml3.dll
0x23a0cd0 \WINDOWS\WinSxS\x86_Microsoft.Windows.Common-Controls_6595b64144ccf1df_6.0.2600.6028_x-ww_61e65202
0x23a1aa0 \Program Files\MSN Gaming Zone\Windows\shvlzm.exe
0x23a1c28 \Documents and Settings\donny\Desktop\PIL-1.1.7.win32-py2.7.exe
0x23a2330 \WINDOWS\system32\sol.exe
0x23a2aa0 \WINDOWS\system32\usmt\migwiz.exe
0x23a2bc0 \Documents and Settings\All Users\Start Menu\Programs\Administrative Tools\Component Services.lnk
0x23a2c90 \Program Files\Windows NT\Accessories\wordpad.exe
0x23a72f8 \Directory
0x23a7458 \WINDOWS\system32\ieframe.dll
0x23a7558 \WINDOWS\system32\advapi32.dll
0x23aa0c0 \WINDOWS\system32\autochk.exe
0x23aa360 \WINDOWS\system32\baserv.dll
0x23aa668 \WINDOWS\system32\winsrv.dll
0x23aac88 \WINDOWS\system32\apphelp.dll
0x23aad0 \Directory
0x23aae58 \WINDOWS\system32\normaliz.dll
0x23cd490 \WINDOWS\system32\mlang.dll
0x23ce268 \WINDOWS\system32\normaliz.dll
0x23ce300 \Directory
0x23ce698 \WINDOWS\system32\imagehlp.dll
0x23ceb60 \LogFile
0x23cec88 \Directory

```

- **Registry Hive List**

Lists all registry hives loaded in memory like SAM and SYSTEM which are needed for password extraction.

Registry hive list shows all loaded hives including SAM, SECURITY, SOFTWARE, SYSTEM, and user-specific NTUSER.DAT files for users donny, LocalService, and NetworkService.

```

(root@kali)~/home/kali/Desktop/Practical Investigation Memory Files_1615939758250 (1)/Practical Investigation Memory Files_1615939758250 (1)
python3 /home/kali/volatility3/vol.py -f Investigation-2.raw windows.registry.hivelist
Volatility 3 Framework 2.28.1
Progress: 100.00 PDB scanning finished
Offset FileFullPath File output
0xe1b179e8 \Device\HarddiskVolume1\Documents and Settings\donny\Local Settings\Application Data\Microsoft\Windows\UsrClass.dat Disabled
0xe1af2b60 \Device\HarddiskVolume1\Documents and Settings\donny\NTUSER.DAT Disabled
0xe17833a0 \Device\HarddiskVolume1\Documents and Settings\LocalService\Local Settings\Application Data\Microsoft\Windows\UsrClass.dat Disabled
0xe18e1b60 \Device\HarddiskVolume1\Documents and Settings\LocalService\NTUSER.DAT Disabled
0xe1763008 \Device\HarddiskVolume1\Documents and Settings\NetworkService\Local Settings\Application Data\Microsoft\Windows\UsrClass.dat Disabled
0xe175cb60 \Device\HarddiskVolume1\Documents and Settings\NetworkService\NTUSER.DAT Disabled
0xe1490b60 \Device\HarddiskVolume1\WINDOWS\system32\config\software Disabled
0xe14a2758 \Device\HarddiskVolume1\WINDOWS\system32\config\default Disabled
0xe1485b60 \Device\HarddiskVolume1\WINDOWS\system32\config\SAM Disabled
0xe1479b60 \Device\HarddiskVolume1\WINDOWS\system32\config\SECURITY Disabled
0xe138b008 Disabled
0xe1035b60 \Device\HarddiskVolume1\WINDOWS\system32\config\system Disabled
0xe102e008 Disabled
(root@kali)~/home/kali/Desktop/Practical Investigation Memory Files_1615939758250 (1)/Practical Investigation Memory Files_1615939758250 (1)

```

- **Handles**

Lists all open handles like files, registry keys and mutexes per process to identify specific malware families.

Mutant (Mutex) Handles

```

(root@kali)~/home/kali/Desktop/Practical Investigation Memory Files_1615939758250 (1)/Practical Investigation Memory Files_1615939758250 (1)
python3 /home/kali/volatility3/vol.py -f Investigation-2.raw windows.handles --pid 620 | grep -i "Mutant"
620 gresswinlogon.exe 0x822147908 scan0x34 finMutant 0x1f0001 userenv: machine policy mutex
620 winlogon.exe 0x82214740 0x38 Mutant 0x1f0001 userenv: Machine Registry policy mutex
620 winlogon.exe 0x8216ea80 0x4c Mutant 0x1f0001 userenv: user policy mutex
620 winlogon.exe 0x8216ea30 0x50 Mutant 0x1f0001 userenv: User Registry policy mutex
620 winlogon.exe 0x8216e618 0x68 Mutant 0x1f0001 -
620 winlogon.exe 0x8216e5d8 0x70 Mutant 0x1f0001 -
620 winlogon.exe 0x8216e598 0x78 Mutant 0x1f0001 -
620 winlogon.exe 0x8216e558 0x7c Mutant 0x1f0001 -
620 winlogon.exe 0x82160580 0x84 Mutant 0x1f0001 -
620 winlogon.exe 0x821b13c8 0xa4 Mutant 0x1f0001 winlogon: Logon UserProfileMapping Mutex
620 winlogon.exe 0x8227e258 0xc8 Mutant 0x1f0001 SingleSesMutex
620 winlogon.exe 0x822cc940 0x104 Mutant 0x1f0001 WPA_PR_MUTEX
620 winlogon.exe 0x821883e8 0x15c Mutant 0x1f0001 ShimCacheMutex
620 winlogon.exe 0x8217f180 0x1e0 Mutant 0x1f0001 msgina: InteractiveLogonMutex
620 winlogon.exe 0x82197180 0x1e4 Mutant 0x1f0001 msgina: InteractiveLogonRequestMutex
620 winlogon.exe 0x821b05d0 0x2c8 Mutant 0x1f0001 -
620 winlogon.exe 0x821c7768 0x2fc Mutant 0x1f0001 -
620 winlogon.exe 0x82089680 0x3c0 Mutant 0x1f0001 -
620 winlogon.exe 0x82087030 0x3c4 Mutant 0x1f0001 WPA_RT_MUTEX
620 winlogon.exe 0x82087c20 0x3c8 Mutant 0x1f0001 WPA_LT_MUTEX
620 winlogon.exe 0x82087b40 0x3cc Mutant 0x1f0001 WPA_HWID_MUTEX
620 winlogon.exe 0x82087b80 0x3d0 Mutant 0x1f0001 WPA_LICSTORE_MUTEX
620 winlogon.exe 0x8215c360 0x7c4 Mutant 0x1f0001 -
620 winlogon.exe 0x822eddb8 0x848 Mutant 0x1f0001 MidiMapper_modLongMessage_RefCnt
620 winlogon.exe 0x8202f800 0x84c Mutant 0x1f0001 MidiMapper_Configure
620 winlogon.exe 0x820346c8 0x850 Mutant 0x1f0001 -

```

- **Privileges**

Shows privileges assigned to each process to detect escalation attempts like SeDebugPrivilege being enabled.

Privileges for the System process (PID 4) and smss.exe (PID 348) are listed. Critical privileges like SeDebugPrivilege, SeImpersonatePrivilege, and SeTcbPrivilege are shown as Present and Enabled.

```

(root@kali) ~ - [~/home/kali/Desktop/Practical Investigation Memory Files_1615939758250 (1)/Practical Investigation Memory Files_1615939758250 (1)]
$ python3 /home/kali/volatility3/vol.py -f Investigation-2.raw windows.privileges
Volatility 3 Framework 2.28.1
Progress: 100.00
PDB scanning finished
PID Process Value Privilege Attributes Description
4 System 7 SeTcbPrivilege Present,Enabled,Default Act as part of the operating system
4 System 2 SeCreateTokenPrivilege Present Create a token object
4 System 9 SeTakeOwnershipPrivilege Present Take ownership of files/objects
4 System 15 SeCreatePagefilePrivilege Present,Enabled,Default Create a pagefile
4 System 4 SeLockMemoryPrivilege Present,Enabled,Default Lock pages in memory
4 System 3 SeAssignPrimaryTokenPrivilege Present Replace a process-level token
4 System 5 SeIncreaseQuotaPrivilege Present Increase quotas
4 System 14 SeIncreaseBasePriorityPrivilege Present,Enabled,Default Increase scheduling priority
4 System 16 SeCreatePermanentPrivilege Present,Enabled,Default Create permanent shared objects
4 System 20 SeDebugPrivilege Present,Enabled,Default Debug programs
4 System 21 SeAuditPrivilege Present,Enabled,Default Generate security audits
4 System 8 SeSecurityPrivilege Present Manage auditing and security log
4 System 22 SeSystemEnvironmentPrivilege Present Edit firmware environment values
4 System 23 SeChangeNotifyPrivilege Present,Enabled,Default Receive notifications of changes to files or directories
4 System 17 SeBackupPrivilege Present Backup files and directories
4 System 18 SeRestorePrivilege Present Restore files and directories
4 System 19 SeShutdownPrivilege Present Shut down the system
4 System 10 SeLoadDriverPrivilege Present Load and unload device drivers
4 System 13 SeProfileSingleProcessPrivilege Present,Enabled,Default Profile a single process
4 System 12 SeSystemTimePrivilege Present Change the system time
4 System 25 SeUndockPrivilege Present Remove computer from docking station
4 System 28 SeManageVolumePrivilege Present Manage the files on a volume
4 System 29 SeImpersonatePrivilege Present,Enabled,Default Impersonate a client after authentication
4 System 30 SeCreateGlobalPrivilege Present,Enabled,Default Create global objects
348 smss.exe 7 SeTcbPrivilege Present,Enabled,Default Act as part of the operating system
348 smss.exe 2 SeCreateTokenPrivilege Present Create a token object
348 smss.exe 9 SeTakeOwnershipPrivilege Present Take ownership of files/objects
348 smss.exe 15 SeCreatePagefilePrivilege Present,Enabled,Default Create a pagefile
348 smss.exe 4 SeLockMemoryPrivilege Present,Enabled,Default Lock pages in memory

```

Privileges for wuaclt.exe (PID 1768), alg.exe (PID 544), and wscntfy.exe (PID 1168) are displayed. wuaclt.exe having SeImpersonatePrivilege enabled is suspicious as it may be a WannaCry-impersonated Windows Update process.

```

1768 wuaclt.exe 22 SeSystemEnvironmentPrivilege Present Edit firmware environment values
1768 wuaclt.exe 23 SeChangeNotifyPrivilege Present,Enabled,Default Receive notifications of changes to files or directories
1768 wuaclt.exe 17 SeBackupPrivilege Present Backup files and directories
1768 wuaclt.exe 18 SeRestorePrivilege Present Restore files and directories
1768 wuaclt.exe 19 SeShutdownPrivilege Present Shut down the system
1768 wuaclt.exe 10 SeLoadDriverPrivilege Present,Enabled Load and unload device drivers
1768 wuaclt.exe 13 SeProfileSingleProcessPrivilege Present,Enabled,Default Profile a single process
1768 wuaclt.exe 12 SeSystemTimePrivilege Present,Enabled Change the system time
1768 wuaclt.exe 25 SeUndockPrivilege Present,Enabled Remove computer from docking station
1768 wuaclt.exe 28 SeManageVolumePrivilege Present Manage the files on a volume
1768 wuaclt.exe 29 SeImpersonatePrivilege Present,Enabled,Default Impersonate a client after authentication
1768 wuaclt.exe 30 SeCreateGlobalPrivilege Present,Enabled,Default Create global objects
544 alg.exe 21 SeAuditPrivilege Present Generate security audits
544 alg.exe 5 SeIncreaseQuotaPrivilege Present Increase quotas
544 alg.exe 3 SeAssignPrimaryTokenPrivilege Present Replace a process-level token
544 alg.exe 23 SeChangeNotifyPrivilege Present,Enabled,Default Receive notifications of changes to files or directories
544 alg.exe 19 SeShutdownPrivilege Present Shut down the system
544 alg.exe 25 SeUndockPrivilege Present Remove computer from docking station
544 alg.exe 29 SeImpersonatePrivilege Present,Enabled,Default Impersonate a client after authentication
544 alg.exe 30 SeCreateGlobalPrivilege Present,Enabled,Default Create global objects
1168 wscntfy.exe 23 SeChangeNotifyPrivilege Present,Enabled,Default Receive notifications of changes to files or directories
1168 wscntfy.exe 8 SeSecurityPrivilege Present Manage auditing and security log
1168 wscntfy.exe 17 SeBackupPrivilege Present Backup files and directories
1168 wscntfy.exe 18 SeRestorePrivilege Present Restore files and directories
1168 wscntfy.exe 12 SeSystemTimePrivilege Present Change the system time
1168 wscntfy.exe 19 SeShutdownPrivilege Present Shut down the system
1168 wscntfy.exe 24 SeRemoteShutdownPrivilege Present Force shutdown from a remote system
1168 wscntfy.exe 9 SeTakeOwnershipPrivilege Present Take ownership of files/objects
1168 wscntfy.exe 20 SeDebugPrivilege Present Debug programs
1168 wscntfy.exe 22 SeSystemEnvironmentPrivilege Present Edit firmware environment values
1168 wscntfy.exe 11 SeSystemProfilePrivilege Present Profile system performance
1168 wscntfy.exe 13 SeProfileSingleProcessPrivilege Present Profile a single process
1168 wscntfy.exe 14 SeIncreaseBasePriorityPrivilege Present Increase scheduling priority
1168 wscntfy.exe 10 SeLoadDriverPrivilege Present Load and unload device drivers
1168 wscntfy.exe 15 SeCreatePagefilePrivilege Present Create a pagefile
1168 wscntfy.exe 5 SeIncreaseQuotaPrivilege Present Increase quotas
1168 wscntfy.exe 25 SeUndockPrivilege Present Remove computer from docking station
1168 wscntfy.exe 28 SeManageVolumePrivilege Present Manage the files on a volume
1168 wscntfy.exe 29 SeImpersonatePrivilege Present,Enabled,Default Impersonate a client after authentication
1168 wscntfy.exe 30 SeCreateGlobalPrivilege Present,Enabled,Default Create global objects

```

